chapter five

DISCRETE RANDOM VARIABLES AND THEIR PROBABILITY DISTRIBUTIONS

They are called wise
who put things in their right order.
– St. Thomas Aquinas

INTRODUCTION

In this chapter, we bring together many of the ideas developed in the first four chapters. We will again work with means and standard deviations and the shapes of distributions. But this time, we will be working with values from the population itself. The populations will be defined using ideas we first saw in the chapter on probability.

We start with the idea of a **random variable** that assigns a number to the outcomes of an experiment. The random variable is classified as either a **continuous random variable** or a **discrete random variable**. The probabilities related to the outcomes are then combined with the random variable to create a **probability distribution**.

Specifically, we will work with **discrete probability distributions** in this chapter. This is the first part of our work with probability distributions that will continue into the next chapter where we study continuous probability distributions that will be used to do what we will call a hypothesis test in inferential statistics.

In the earlier chapters we looked at collections of sample data. We then calculated sample statistics as estimates of the population parameters, and we were concerned that we had representative samples. Now we have an opportunity to look at the population itself through the probability distribution. Given a probability distribution, it will now be possible to find the **population mean**, and the **population standard deviation**. A very useful concept called the **expected value** of a random variable will be defined as the population mean.

The graphs that we created in the earlier chapters provided us an idea about the shape of the distribution of the population. In this chapter we create a **probability histogram** using bars to represent the probability for each value of the random variable. The probability histogram, like the grouped frequency distribution histogram, provides insight into the shape of the population defined by the probability distribution.

We will look at three important, and useful, discrete probability distributions. The first is named the **binomial probability distribution**. It involves a **binomial experiment** with repeated independent trials resulting in one of two possible outcomes, success or failure. When the trials are not independent, we use the **hypergeometric probability distribution**. The third, the **Poisson probability distribution**, involves independent occurrences of an event in a fixed interval. These three distributions are involved in a large number of applications.

This chapter is important to what follows in that we learn how to find the probability of getting specific values of the random variable. In particular, we relate the probabilities to the area in the bars of the probability histogram. We are interested in events that are rare; events that have less than a 5% chance of occurring. When such events do occur, we consider them significant. These ideas will be used in hypothesis testing in later chapters.

CHAPTER 5 HOT SPOTS

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If you find other HOT SPOTS, write them down and use them as a focus of your discussions in the study group. Or you can use the HOT SPOT as the topic for a help session with your professor.

USE THIS PAGE FOR KEEPING TRACK OF TERMS AND NOTATION

Hot Spot #1 – Discrete Versus Continuous Random Variables

The definitions for both discrete and continuous random variables are rather difficult to understand at first. There is a fairly simple way of thinking about discrete and continuous random variables. If the variable represents something that is counted, then the random variable is discrete. If the variable represents something that is measured, then the random variable is continuous.

It is not enough to just look at the values given for the random variable. For example, if the random variable represents the time to do a statistics problem on a test, we might simply give the time to the nearest five minutes. Even though the time is reported as an integer, the measurement of time could have been any value in an interval of 0 to 60 minutes. Hence, the random variable in this case would be continuous.

It is much easier to determine when we are counting. For example, in this chapter we will count the number of successes in the repeated trials of an experiment. The result is a discrete random variable.

Again, to determine whether the random variable is discrete or continuous, it is best to simply ask if we counted or measured to get the value.

COUNT ⟶ DISCRETE

MEASURE ⟶ CONTINUOUS

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MANN Sections 5.1.1 And 5.1.2

See the problem set in sections 5.1.1 and 5.1.2 for examples of both discrete and continuous variables.

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Mann
Section 5.2

Hot Spot #2 – Checking For A Probability Distribution

It is easy to check if we have a probability distribution. We must satisfy the following two conditions:

$$0 \le P(x) \le 1 \text{ for all } x$$
$$\sum P(x) = 1$$

The first condition states that the probability of x must be between 0 and 1 for all values of the random variable. The second condition states that the sum of the probabilities for all possible values of the random variable must equal 1. The following example illustrates what should be done to check on a potential probability distribution. Determine if the following function is a probability distribution:

$$P(x) = \frac{x}{5} \text{ for } x = 1,2,3,4,5$$

We first check to see if $P(x)$ is between 0 and 1 for all x.

$$P(1) = \tfrac{1}{5}$$
$$P(2) = \tfrac{2}{5}$$
$$P(3) = \tfrac{3}{5}$$
$$P(4) = \tfrac{4}{5}$$
$$P(5) = \tfrac{5}{5}$$

The function satisfies the first check. Now we need to find the sum of all the probabilities.

$$P(1) + P(2) + P(3) + P(4) + P(5) = \tfrac{1}{5} + \tfrac{2}{5} + \tfrac{3}{5} + \tfrac{4}{5} + \tfrac{5}{5}$$
$$\sum P(x) = 3$$

The function fails the second check.

Hot Spot #2 Sample Problem and Solution: Determine if the following function is a probability distribution function:

$$P(x) = \frac{x^2}{3} \text{ for } x = 1,2,3$$

Solution: $P(x) > 1$ for $x = 2$ and $x = 3$.

This is not a probability distribution function.

Sample Problem and Answer: Determine if the following table is a probability distribution:

x	$P(x)$
0	0
1	0.1
2	0.4
3	0.9

Answer: The sum of the probabilities is more than 1.

This is not a probability distribution.

Sample Problem and Answer: Determine if the following is a probability distribution:

x	0	1	2	3
$P(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Answer: Each probability is between 0 and 1, and the sum of the probabilities is 1; therefore, this is a probability distribution.

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Mann
Section 5.6.1

Hot Spot #3 – Determining If We Have A Binomial Random Variable

Deciding if the random variable is a binomial random variable at first seems like one of the more difficult things to do in this chapter. However, there are only three requirements we need to check.

The first requirement is that the problem involves repeated trials. This is simply a statement that the same thing is done each time. The problems often involve tossing a die repeatedly or flipping a coin repeatedly; but can involve any repetitious task such as doing test questions, shooting baskets, etc.

The second requirement is that each trial is identical and results in one of two possible outcomes. One of the outcomes will be considered a success, the other a failure. The success must be the same in each trial. For example, if we toss a die and decide that getting a 3 is a success, we can not change our mind after 5 tosses and decide that getting a 6 is a success.

The third requirement is that the probability must be the same for each trial. This is not a problem when we toss dice or flip coins. It may indeed be a problem when we shoot baskets or do test questions. In such cases, we may need to make an assumption that the probability of a success does not change from trial to trial.

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Mann
Section 5.6.2

Hot Spot #4 – Using $P(x) = {}_nC_x p^x q^{n-x}$ For Binomial Random Variables

There is an important pattern to observe when using the binomial formula to calculate probabilities for a binomial random variable. The exponent x used on p, the probability of success, is the same x in $P(x)$. For example, if $n = 5$ and $p = 0.2$, then $P(3)$ is written as follows:

$$P(3) = {}_5C_3(0.2)^3(0.8)^2$$

We need to notice where the 3's are placed. We have the first 3 in $P(3)$. The second 3 is in ${}_5C_3$. The last 3 is in $(0.2)^3$. This pattern is always used when we use the binomial probability formula to find $P(x)$.

Hot Spot #4 Sample Problem and Solution: Given x has a binomial distribution with $n = 4$ and $p = 0.3$, write the expression for $P(3)$.

Solution: Step 1. The 3 in $P(3)$ is the key number. We write ${}_nC_x$ as ${}_4C_3$ and p^x as p^3.

Step 2. The answer will be in the form ${}_4C_3(0.3)^3(0.7)^1$

Sample Problem and Answer: Given the binomial random variable with $n = 8$ and $p = 0.4$, write the expression for $P(5)$.

Answer: The answer will be in the form ${}_8C_5(0.4)^5(0.6)^3$

Sample Problem and Answer: Given the binomial distribution with $n = 5$ and $p = \frac{1}{3}$, write the expression for $P(4)$.

Answer: The answer will be in the form ${}_5C_4\left(\frac{1}{3}\right)^4\left(\frac{2}{3}\right)^1$

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Mann Sections 5.5.1, 5.5.2, 5.6.2

Hot Spot #5 – Computing ${}_nC_x = \frac{n!}{x!(n-x)!}$ Using Pascal's Triangle

The expression $\frac{n!}{x!(n-x)!}$ is used in the formula for $P(x)$ for a binomial random variable. Another way of writing the expression is as what we call a combinatorial. We use the notation ${}_nC_x$ to represent $\frac{n!}{x!(n-x)!}$. The notation ${}_nC_x$ is used when we want to know how many different combinations are possible when we take n items x at a time.

For example, if we have 5 items such as the letters a,b,c,d and e, and want to take 3 letters at a time, we would write ${}_5C_3$. If we only need to do one computation, the formula using factorials is not that difficult to use. But if we need to do several calculations, as would be necessary in creating a binomial probability distribution table, there is an easier way to find the values of ${}_nC_x$.

The computation can be made very easy if we use Pascal's triangle. The triangle is created by looking at the coefficients of the binomial expansion $(p+q)^n$ for $n = 0,1,2,3,4,,$ etc. The first four expansions are as follows:

$$(p+q)^0 = 1$$
$$(p+q)^1 = 1p^1 + 1q^1$$
$$(p+q)^2 = 1p^2 + 2p^1q^1 + 1q^2$$
$$(p+q)^3 = 1p^3 + 3p^2q^1 + 3p^1q^2 + 1q^3$$

We create Pascal's triangle by writing the coefficients of the terms in the following form:

Figure 5–1: Pascal's Triangle

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 1 & & 1 & & \\ & 1 & & 2 & & 1 & \\ 1 & & 3 & & 3 & & 1 \end{array}$$

We get the next row of the triangle by adding the numbers in the row with 1 3 3 1 as follows:

$$\begin{array}{ccccccccc} & 1 & & 3 & & 3 & & 1 & \\ 1 & & 4 & & 6 & & 4 & & 1 \end{array}$$

This new row gives the coefficients of the terms in $(p+q)^4$:

$$(p+q)^4 = 1p^4 + 4p^3q^1 + 6p^2q^2 + 4p^1q^3 + 1q^4$$

We get the next row of the triangle by adding the numbers in the row with 1 4 6 4 1 as follows:

$$\begin{array}{ccccccccccc} & 1 & & 4 & & 6 & & 4 & & 1 & \\ 1 & & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$

This new row gives the coefficients of the terms in $(p+q)^5$:

$$(p+q)^5 = 1p^5 + 5p^4q^1 + 10p^3q^2 + 10p^2q^3 + 5p^1q^4 + 1q^5$$

We can continue expanding the triangle to as many rows as we need.

It is important to study the pattern of the exponents in the expansions of $(p+q)^n$. We first need to observe that the sum of the exponents for each term is always equal to the value of n. Next, as we examine the terms going from left to right, the exponent on p goes down 1 each time as the exponent on q goes up 1 each time. Again, we will need to focus on the value of x in the formula $P(x) = {}_nC_x p^x q^{n-x}$.

Now we can use Pascal's Triangle to find $P(x)$ for a binomial random variable. The only difficulty is to decide which term of the triangle to use. All we need to do is be aware of the pattern in the terms of the expansion. If we need to find $P(x)$, the probability of x for a particular problem, we will know n and get x from the notation $P(x)$. The value of n tells us to use the $n+1$ row of the triangle. The value of x gives us the exponent we will use on p for the appropriate term of the expansion. The following problems illustrate the process.

Hot Spot #5 Sample Problem and Solution: Given the binomial distribution with $n = 3$ and $p = 0.1$, find $P(2)$.

Solution: Step 1. $n = 3$, so we want the row of the triangle for $(p+q)^3$.

Step 2. $P(2)$ tells us $x = 2$, so we want the term ${}_3C_2p^2q^1$.

In Mann The Answers Are Written In The Decimal Form.

Step 3. Substituting for p and q we get $P(2) = {}_3C_2(0.1)^2(0.9)^1$.

Step 4. We go to Pascal's triangle to the row 1 3 3 1 and use the first 3 for the value of ${}_3C_2$.

Step 5. We can either leave the answer in the form $P(2) = 3(0.1)^2(0.9)^1$, or perform the computation to get $P(2) = 0.027$.

Sample Problem and Answer: Given the binomial distribution with $n = 4$ and $p = 0.3$, find $P(1)$.

Answer: Step 1. $n = 4$, so we want the row of the triangle for $(p+q)^4$.

Step 2. $P(1)$ tells us $x = 1$, so we want the term ${}_4C_1 p^1 q^3$.

Step 3. Substituting for p and q we get $P(1) = {}_4C_1(0.3)^1(0.7)^3$.

Step 4. We go to Pascal's triangle to the row 1 4 6 4 1 and use the second 4 for the value of ${}_4C_1$.

Step 5. We can either leave the answer in the form $P(1) = 4(0.3)^1(0.7)^3$ or perform the computation to get $P(1) = 0.4116$.

Sample Problem and Answer: Given the binomial random variable with $n = 5$ and $p = \frac{1}{4}$, use Pascal's triangle to find $P(x)$ for $x = 5,4,3,2,1,0$.

Answer:

x	$P(x)$
5	$1\left(\frac{1}{4}\right)^5 = 0.0010$
4	$5\left(\frac{1}{4}\right)^4\left(\frac{3}{4}\right)^1 = 0.0146$
3	$10\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^2 = 0.0879$
2	$10\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^3 = 0.2637$
1	$5\left(\frac{1}{4}\right)^1\left(\frac{3}{4}\right)^4 = 0.3955$
0	$1\left(\frac{3}{4}\right)^5 = 0.2373$

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Mann
Section 5.6.5

Hot Spot #6 – When To Use $\mu = np$ And $\sigma^2 = npq$

The general formulas for the mean and variance of a discrete probability distribution are:

$$\mu = \sum xP(x) \text{ and } \sigma^2 = \sum (x-\mu)^2 P(x)$$

The formulas will work for any discrete probability distribution, but they are time consuming to use. Fortunately, we can use an easier and faster formula when we have a binomial random variable.

The formulas for the mean and variance of a binomial random variable are:

$$\mu = np \text{ and } \sigma^2 = npq$$

Remember, these shortcut formulas can only be used if the random variable is from a binomial experiment. If the formulas are used for a non-binomial discrete probability distribution, we will get an incorrect answer. The following example shows what happens when we use the wrong formula.

The following function is a probability distribution:

x	1	2	3	4	5	6
$P(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

We can use the formula $\mu = \sum xP(x)$ to find the mean because this formula will give us the mean for any discrete probability distribution.

$$\mu = 1\left(\tfrac{1}{6}\right) + 2\left(\tfrac{1}{6}\right) + 3\left(\tfrac{1}{6}\right) + 4\left(\tfrac{1}{6}\right) + 5\left(\tfrac{1}{6}\right) + 6\left(\tfrac{1}{6}\right)$$

$$\mu = (1+2+3+4+5+6)\left(\tfrac{1}{6}\right) = \tfrac{21}{6} = 3.5$$

Now, we will make the faulty assumption that we have a binomial distribution and use the wrong formula for the problem.

$$\mu = np = 6\left(\tfrac{1}{6}\right) = 1$$

In this problem, it is very clear that the distribution is not a binomial. Furthermore, it is easy to see that the distribution is not centered at $\mu = 1$. Hopefully, we would see our mistake if we used the wrong formula. Of course, it is best to avoid the mistake in the first place and use the correct formula. When in doubt, do not use the shortcut formula.

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Mann Sections 5.6.3 And 5.7.1

Hot Spot #7 – Interpreting The Probability Of At Least k Successes

The phrase, "find the probability of at least k successes", often causes some difficulty. Part of the problem is to understand what the phrase, *at least*, means. For example, *at least* 2 means 2 or more. So if $n = 5$ in a discrete probability distribution, the phrase *at least 2* would mean that we are interested in 2, 3, 4, or 5 successes.

The probability of at least 2 successes would be written as $P(x \geq 2)$ and include $P(x = 2) + P(x = 3) + P(x = 4) + P(x = 5)$. The idea of a complement from the probability chapter could be used to simplify the problem. We would write $P(x \geq 2)$ as $1 - P(x \leq 1)$. Then, we would only have to find $P(x = 0) + P(x = 1)$ and subtract the sum from 1.

Hot Spot #7 Sample Problem and Solution: Given the binomial distribution with $n = 5$ and $p = 0.4$, find the probability of at least one success.

Solution: Step 1. We want to find $P(x \geq 1)$. We use the complement to write the following:

$$P(x \geq 1) = 1 - P(x < 1) = 1 - P(x = 0)$$

Step 2. $P(x = 0) = {}_5C_0(0.4)^0(0.6)^5 = 0.07776$

Step 3. $P(x \geq 1) = 1 - 0.07776 = 0.92224$

Sample Problem and Answer: Given the binomial distribution with $n = 4$ and $p = 0.2$, find the probability of at least 2 successes.

Answer: $P(x \geq 2) = 1 - P(x \leq 1) = 1 - [P(x = 1) + P(x = 0)]$

$$= 1 - \left[4(0.2)^1(0.8)^3 + 1(0.8)^4\right]$$

$$= 1 - (0.4096 + 0.4096) = 1 - 0.8192 = 0.1808$$

✷ *Mann Section 5.6.3*

Sample Problem and Answer: Given the Poisson distribution with $n = 20$ and $\lambda = 0.5$, find the probability of at least one success.

Answer: $P(x \geq 1) = 1 - P(x \leq 0) = 1 - \dfrac{(0.5)^0 e^{-.5}}{0!} = 1 - 0.6065 = 0.3935$

Hot Spot #8 – Binomial Probability Distribution Tables

When we work with a binomial random variable, there are two common options that are used to find the probability for a specific value of the random variable. The easiest method is to use either a computer or a calculator. When we input a value of n and p, the calculator or computer will give us the probability we want.

The second method is to use a table of probabilities. Binomial probability tables are readily available for small values of n and common values of p, such as 0.1, 0.2, 0.3, etc. Later on we will work with a method that gives approximate answers for large values of n. However, there may still be a need to construct tables for value of p not found in a typical set of tables. We can then use the binomial probability formula $P(x) = {}_nC_x p^x q^{n-x}$ to construct a table of binomial probabilities. The easiest way to do the problem is using the appropriate expansion of $(p+q)^n$ as shown in the following sample problems.

See Hot Spot #5 for help with using Pascal's Triangle.

Hot Spot #8 Sample Problem and Solution: Create the probability table for the binomial probability distribution with $n = 3$ and $p = 0.4$.

Solution: Step 1. We want $(p+q)^3 = 1p^3 + 3p^2q^1 + 3p^1q^2 + 1q^3$.

Step 2. We create the table as follows:

x	$P(x)$
3	$1p^3$
2	$3p^2q^1$
1	$3p^1q^2$
0	$1q^3$

Step 3. We now substitute for p and q as follows:

x	$P(x)$
3	$1(0.4)^3$
2	$3(0.4)^2(0.6)^1$
1	$3(0.4)^1(0.6)^2$
0	$1(0.6)^3$

Step 4. We can write the values with decimals as follows:

x	$P(x)$
3	0.064
2	0.288
1	0.432
0	0.216

Sample Problem and Answer: Given the binomial random variable with $n = 2$ and $p = 0.8$, create the binomial probability table.

Answer:

x	$P(x)$
2	$1(0.8)^2$
1	$2(0.8)^1(0.2)^1$
0	$1(0.2)^2$

Sample Problem and Answer: Given the binomial distribution with $n = 5$ and $p = \frac{1}{3}$, create the table of binomial probabilities.

Answer:

x	$P(x)$
5	$1\left(\frac{1}{3}\right)^5$
4	$5\left(\frac{1}{3}\right)^4\left(\frac{2}{3}\right)^1$
3	$10\left(\frac{1}{3}\right)^3\left(\frac{2}{3}\right)^2$
2	$10\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^3$
1	$5\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^4$
0	$1\left(\frac{2}{3}\right)^5$